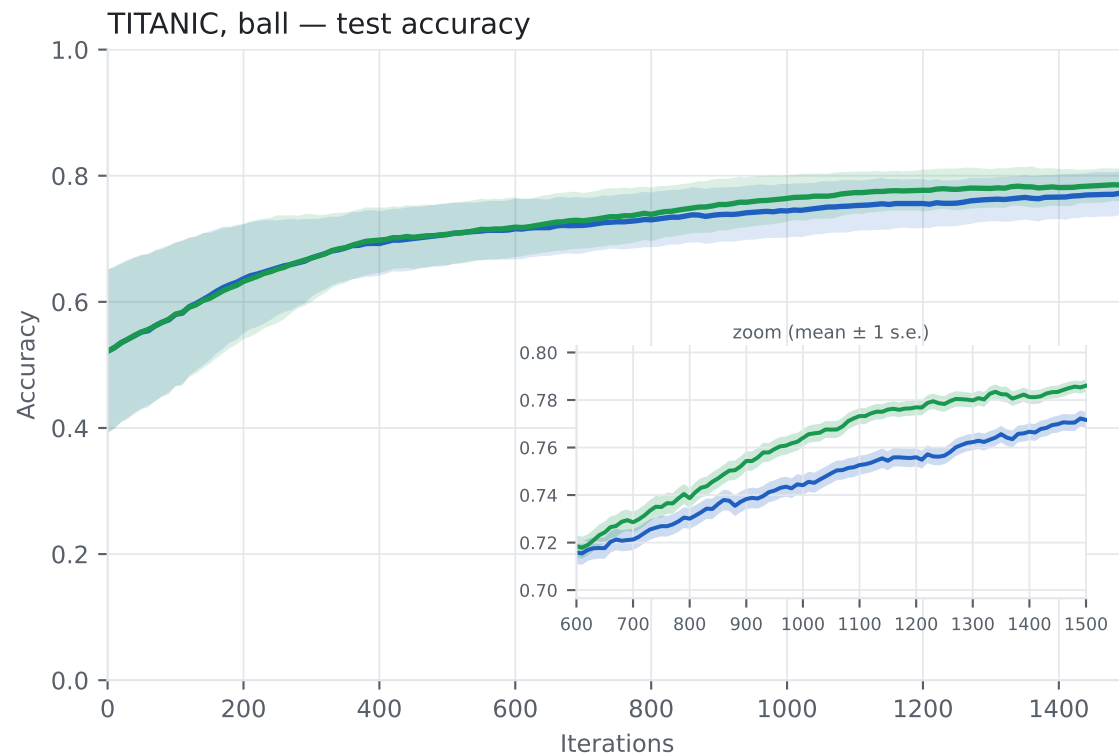
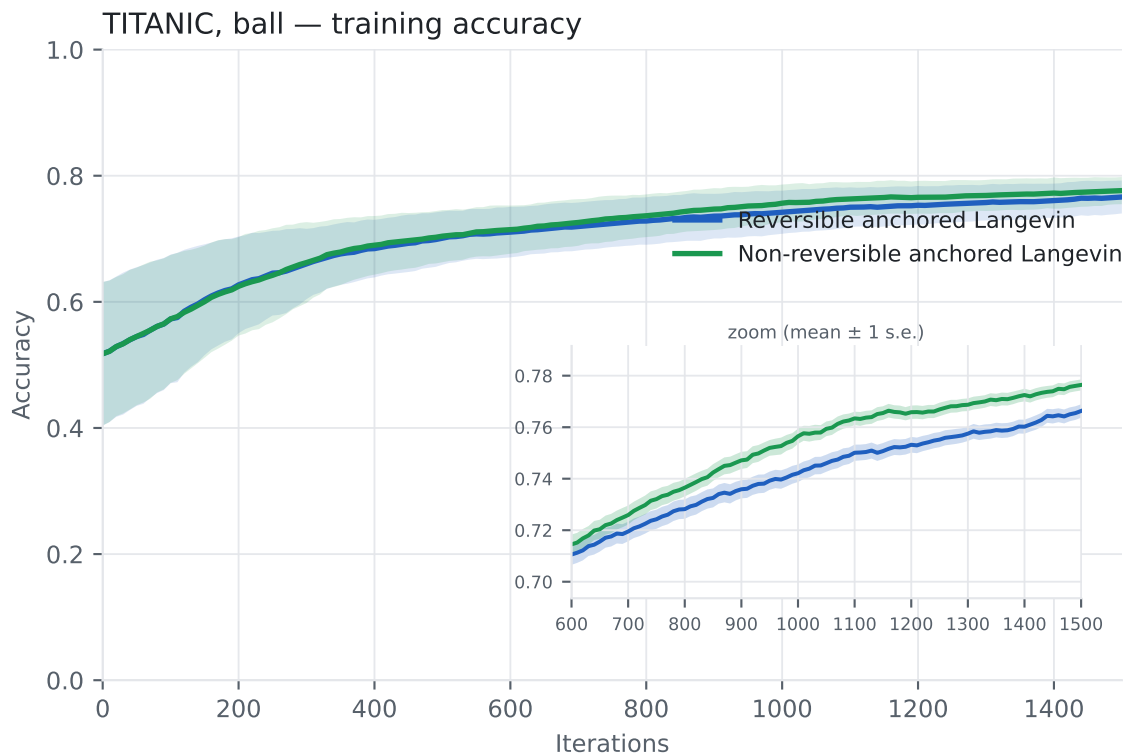


# titanic\_ball: exact gradient, LASSO anchor



TITANIC -- ball  $\|w\|^2 \leq 2$  (radius  $\sqrt{2}$ ).  $d = 10$  (intercept  $w_0$  + nine features), no mini-batching: the EXACT gradient is used at every step.  $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j=1}^9 |w_j|$ ;  $U_0$  replaces  $|w_j|$  by  $\sqrt{w_j^2 + \delta^2}$ ;  $a = \exp(U - U_0)$ .  $\sigma = 10$ ,  $\lambda = 7.12 = 0.01$   $n_{\text{train}}$ ,  $\delta = 0.01082$ , so  $a$  in  $[0.50, 1]$ . Block strengths  $s = (5, 5, 5)$  on  $w_1..w_9$  only. Step  $\eta = 1e-05$ .  $R = 100$  coupled replicates (same  $W_0$ , same Gaussian stream) on one fixed stratified 80/20 split ( $n_{\text{train}} = 712$ ,  $n_{\text{test}} = 179$ ). Paired final-iterate difference, non-reversible minus reversible: training  $+0.0101$  ( $t = +4.09$ ), test  $+0.0144$  ( $t = +4.55$ ). Bands: mean  $\pm$  1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.